

WOMEN SAFETY DEVICE

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Abstract

The Women Safety Device with GPS Tracking and Alert Messaging is a cutting-edge personal safety system for women, offering continuous monitoring and automatic emergency response. It monitors the user's heartbeat and, upon detecting abnormal increase, sends an alert with the user's location to a pre-stored emergency contact. An emergency button allows manual activation, sending a text and initiating a call after 10 seconds, during which the device's microphone transmits surrounding sounds. This automated, hands-free system ensures quick assistance, providing a reliable and efficient solution for women's safety in emergencies.

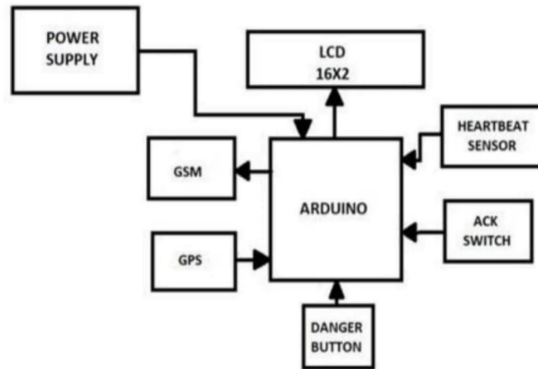
Components Used

1.GPS NEO6MV2 2.Arduino
NANO 3.GSM SIM900A 4.RF
Transceiver 5.LCD Display
6.HT12D IC 7.HT12E IC

Objective

A wearable Women Safety Device with GPS and alert messaging ensures continuous monitoring and swift emergency response. It detects distress, sending location and audio alerts to designated contacts. A manual emergency button and compact design make it easy to use. The device uses GPS, GSM, and heartbeat sensors to provide a safer environment for women in critical situations.

Proposed system -Block Diagram

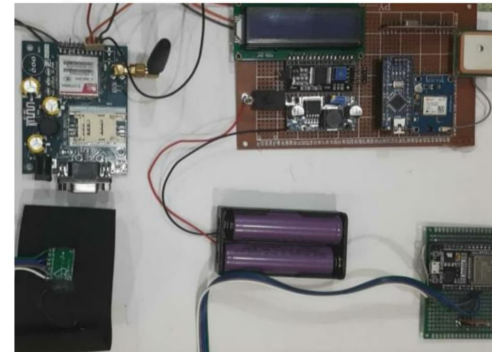


Methodology

Heart beat sensor data obtained is encoded by the encoder and transmitted via transmitter. On the other hand, the data is received at the receiver. The decoder decodes it and displays it on the LCD Screen. If the heartbeat is below 130, the heartbeat sensor keeps on monitoring the pulse-rate. If the heartbeat goes above the specified value i.e. normal heart rate, then the GSM module will send the help me message along with current location obtained from GPS module to the pre-stored contacts. Separately, a Danger button is also there, in case the person feels anything weird then she can directly press it. The message along with the location will be sent to the pre-stored contact. After 10s a call will be sent to contact and once the call is attended nearby and sounds can be heard. When the supply is switched on by connecting the wires, the heartbeat sensor starts monitoring the heartbeat. Location will be displayed on the LCD screen.

Result

The project successfully achieved its objectives: sending a message with the victim's current location to a stored number when the button is pressed and sending a follow-up message indicating the person is safe when the button is released.



Conclusion

The proposed design aims to address recent safety challenges women face by providing a cost-effective, compact, and wearable device using Arduino Nano. This device shares the user's location with trusted contacts and sends text messages. Combining GSM and GPS with Arduino technology, the system enables quick transmission of the victim's message and location during emergencies, aiding in swift identification. However, connectivity issues in some areas remain a challenge. Future work should focus on integrating GPS to alert nearby public locations. While our current model effectively tracks and notifies pre-selected contacts and voice detector tracks the voice.